

2(a)(i)	any two from: 1 no / less, ATP for the, sodium-potassium pump / active transport of Na^+ and K^+ ; 2 reduces / prevents, (re-establishment of) resting potential ; 3 reduces transmission of action potential (along the axon) ;	2
2(a)(ii)	any two from: 1 decreases (speed) / lower (speed than expected) ; 2 in both, nerves / median and peroneal / arm and leg ; 3 processed data quote ; e.g. median mean 43 m s^{-1} / peroneal mean 31.3 m s^{-1} 4 AVP ; e.g. person 1 not affected so much	2
2(a)(iii)	any two from: 1 takes longer for impulses to reach, neuromuscular junction / muscles ; 2 fewer / slower, muscle contractions ; 3 walking problem described ; e.g. reduced muscle control / walks slower / stumbles / trips 4 slower reflexes ;	2

Question	Answer	Marks
2(b)	accept 'neurone activity' for 'balance time' throughout <i>(had benefit is justified because):</i> 1 (mice with therapy) rate of decline of balance time, slower / less steep, from, 18 to 20 months / initially / at the start or (mice with therapy) have a longer balance time ; <i>(not cure is justified since):</i> 2 (mice with therapy and mice with no therapy) have, same / similar, balance time, at 23 months / at the end or mice with therapy have a, downwards trend / decrease in balance time or mice with therapy have shorter balance time (compared to mice without GAN) ; <i>(results are not affected by age is not justified because):</i> 3 mice without GAN increase balance time, during the experiment / with age or balance time decreases for mice with GAN, during the experiment / with age ; 4 comparative figures to support any marking point or ref. to need for statistical test ;	4